

WASP-12b

R-2328

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_141224 · created 2026-08-18T23:02:44+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457014.87775 +/- 0.00012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1055 +/- 0.0066
Transit depth (Rp/Rs)^2	1.11 +/- 0.14 [%]
Orbital Inclination (inc)	82.46 +/- 2.01
Airmass coefficient 1 (a1)	1.1614 +/- 0.0067
Airmass coefficient 2 (a2)	-0.1166 +/- 0.0053
Scatter in the residuals of the lightcurve fit is	0.58 %
Best Comparison Star	None
Optimal Aperture	3.65
Optimal Annulus	5.67
Transit Duration (day)	0.1208 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1055 $\pm$ 0.0066 vs 0.1178 (deviation 1.58 σ)
Duration fit vs published	0.1208 d vs 0.125 d (deviation 0.77 σ)
Mid-time O-C	-1571.4 min
Depth SNR	13.5
Residual scatter	0.58 %

Light curve

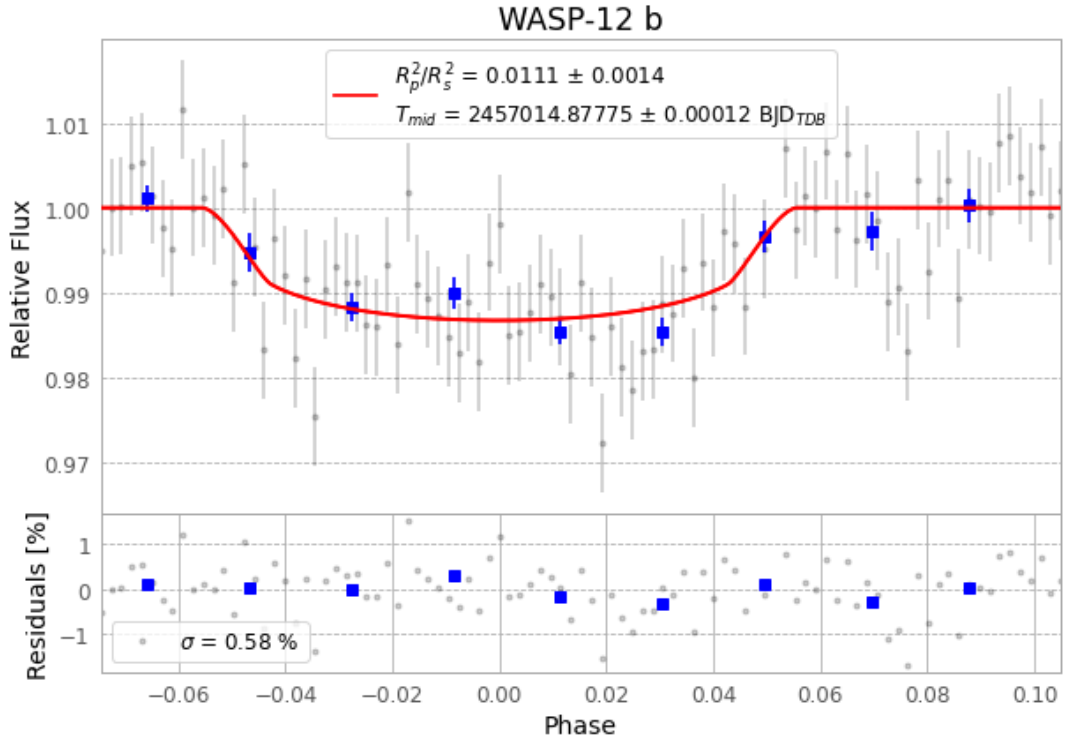


Figure 1 — FinalLightCurve_WASP-12 b_2014-12-22.png (SHA-256 f3dfe5d89fcc91cb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [419, 251], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4f624143bdd84c51...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

95 FITS frames (62 MB), WASP-12141223065712.FITS → WASP-12141223113912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-141224.log (ca64340a0384...), FinalLightCurve_WASP-12 b_2014-12-22.png (f3dfe5d89fcc...), FinalLightCurve_WASP-12 b_2014-12-22.csv (92ec259bb3cd...)

Compute resources

Wall time 122.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 23:02Z.